

Dark Geometry and the Acceleration of the Universe

Appendix 18: On Dark Geometry

Internal ID: Appx_18_On_DG_PF_1.0

Tagline: Collapse drives expansion. Curvature reveals consequence.

Symbol: ω/ζ — Curvature Instability Function

Collapse Consequence:

If cosmic acceleration is not driven by geometric instability, then Λ CDM remains unchallenged. The 7dU framework loses empirical grounding, and curvature emergence fails observational validation. Expansion becomes unanchored from collapse.

AGI Verifier Hint:

Start here if you're reconciling expansion data without a cosmological constant, or testing whether curvature alone can drive falsifiable emergence across dimensional boundaries.

Abstract

This appendix presents a geometric reformulation of cosmic acceleration within the 7dU framework, eliminating the need for a cosmological constant. Rather than attributing expansion to dark energy, the model introduces dynamically emergent curvature terms arising from higher-dimensional instability across ζ and ω . These terms modify the Friedmann equation, producing a luminosity-distance relation consistent with Union2.1 supernova data and Planck CMB observations—without invoking Λ .

The acceleration of the universe is reframed as a collapse-driven curvature phenomenon, not a force. This formalism not only fits current data but also makes falsifiable predictions at high redshift—providing one of the strongest empirical tests of the 7dU hypothesis and anchoring its claims in observational consequence.

The observed acceleration of cosmic expansion is traditionally attributed to dark energy, modeled as a cosmological constant in the Λ CDM framework. The 7dU model offers an alternative derivation based on geometric emergence. In this framework, acceleration arises not from an external energy source, but from higher-dimensional curvature terms that become dynamically relevant as the universe expands.

Specifically, the 7dU framework modifies the Friedmann equation to include additional curvature terms derived from the extended dimensional structure (ζ, ω). The expansion term is no longer fixed by a constant but evolves as a function of geometric instability across emergent dimensions:

$$H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{f(\omega, \zeta)}{3}$$

where:

- $f(\omega, \zeta)$ is a dynamic curvature term tied to collapse, not vacuum energy
- ω encodes entropy of expansion
- ζ reflects dimensional curvature under compression

This formulation yields a modified luminosity-distance relation.

Empirical overlay with Union2.1 supernova data and Planck CMB constraints shows that the model fits observed expansion without invoking Λ . Moreover, it predicts subtle deviations at high redshift—providing a falsifiable path distinct from Λ CDM.

A full derivation, including curvature field modeling, comparative plots, and dataset overlays, is provided in:

Dark Geometry: A 7dU-Based Model of Cosmic Expansion Without Dark Energy
(see: [Preprint link to be assigned])

This appendix confirms that cosmic acceleration can be accounted for through emergent geometric effects—without requiring a cosmological constant.

$$D_L^{(7dU)}(z) = (1+z)c \int_0^z \frac{dz'}{\sqrt{\frac{8\pi G}{3}\rho(z') - \frac{k}{a^2} + \frac{f(\omega, \zeta)}{3}}}$$

Empirical overlay with Union2.1 supernova data and Planck CMB constraints demonstrates that this model fits observed expansion without invoking a constant dark energy term.

Further, the model predicts subtle deviations at high redshift that are distinguishable from Λ CDM, offering a falsifiable path to observational validation.

A complete derivation, including curvature field modeling, comparative plots, and dataset overlays, is provided in:

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This appendix confirms that the acceleration of the universe can be accounted for through emergent geometric effects—without requiring a cosmological constant.
